DVA Capstone 2 : Credit Shield Identifying and Mitigating Loan Default Risk

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Sector : Finance | Group : 3

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GitHub: Link
Tableau: Link

Context & Problem Statement

Sector Context

- This project is in the Finance(credit risk) sector, where loan defaults lead to major financial losses for institutions.
- It matters because accurately predicting risky customers helps reduce losses and improve lending decisions.
- The key decision-makers are credit risk officers, analysts, and portfolio managers.
- They use these insights to adjust credit limits, monitor high-risk customers, and refine lending policies.

Problem Statement

- Identify the key demographic and behavioral factors driving credit default and segment customers by risk.
- Enable proactive actions like adjusting credit limits and monitoring high-risk customers to reduce default rates.

Objective

- Support data-driven decisions to identify high-risk customers and take proactive actions like adjusting creditlimits, refining lending criteria, and increasing monitoring to reduce default rates

Data Engineering: Source to Pipeline

Source Dataset

Default of Credit Card Clients (UCI Machine Learning Repository), 30,000 rows, 25 columns, covering the period April 2005 to September 2005.

Notebooks 01 & 02

Data extraction and cleaning notebooks are committed to GitHub, with a clear summary of all transformations including handling nulls, outliers, and feature engineering steps.

Python ETL

- **Missing Values :** Checked and handled nulls using imputation or removal to ensure data consistency.
- **Outliers**: Detected extreme values using IQR/boxplots and capped or treated them.
- **Standardization**: Created features and scaled key variables for better analysis.

Data Dictionary

Key columns include LIMIT_BAL (credit limit indicating financial capacity), avg_delay (average payment delay as a behavioral risk indicator), utilization_ratio (credit used vs limit showing financial stress), and default_status (target variable indicating whether the customer defaulted).

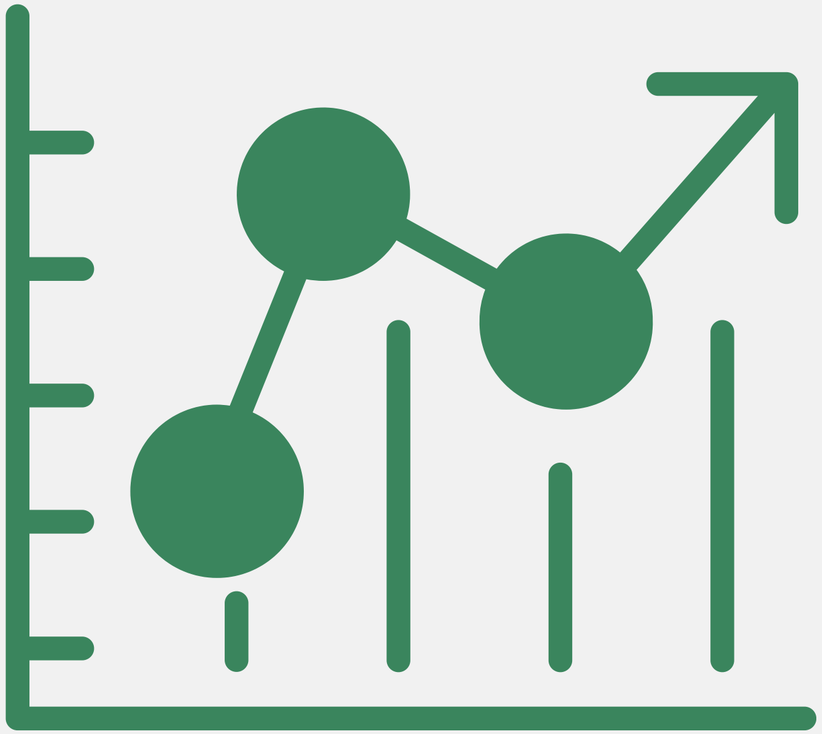
KPI & Metrics Framework

What Are You Measuring?

- Default Rate (%) – proportion of customers who default
- Risk Tier Exposure – average credit limit across risk segments
- Behavioral Delay Index (avg_delay) – average payment delay indicating risk level

Why These KPIs?

- Default Rate shows overall portfolio risk and financial loss.
- Risk Tier Exposure highlights how much credit is at risk in each segment.
- Behavioral Delay Index (avg_delay) identifies early warning signs of default.



How Are They Computed?

- Default Rate (%) = (sum(default_status) / total_customers) * 100 — calculated in notebooks/04_statistical_analysis.ipynb
- Risk Tier Exposure = mean(LIMIT_BAL) grouped by risk_tier — computed in notebooks/05_final_load_prep.ipynb
- Behavioral Delay Index (avg_delay) = mean(PAY_0 to PAY_6) — created during feature engineering in notebooks/02_cleaning.ipynb

Key Insights: Exploratory Data Analysis

High credit utilization customers default more frequently → **Over-reliance on credit indicates financial stress and repayment risk.**

Customers with higher payment delays are significantly more likely to default → **Delayed repayment behavior is the strongest early warning signal of risk.**

A small segment of customers contributes to a large share of defaults → **Risk is concentrated, enabling targeted intervention strategies.**

Customers with lower credit limits show higher default rates → **Financially constrained customers are more vulnerable to default.**

Graduate-level customers show lower default rates compared to other education groups → **Education level influences financial discipline and risk profile.**

Advanced Analysis

New Understanding

- Behavioralmetrics predict default earlier and more accurately than demographics.
- Enabled shift from descriptive analysis to proactive, predictive decision-making.

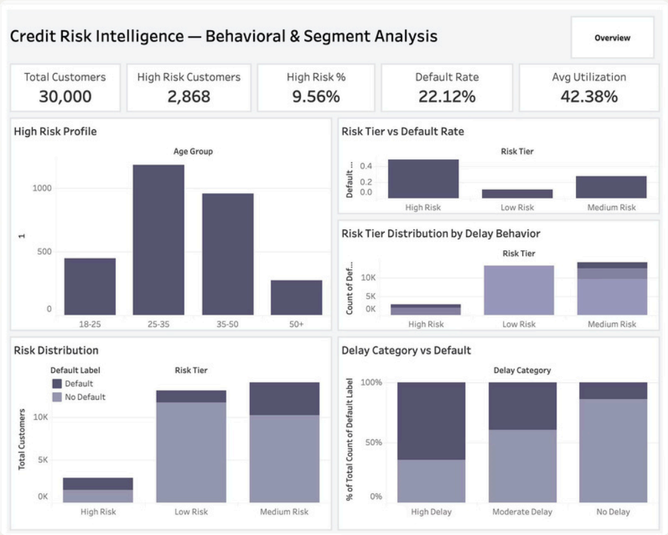
Statistical & Analytical Depth

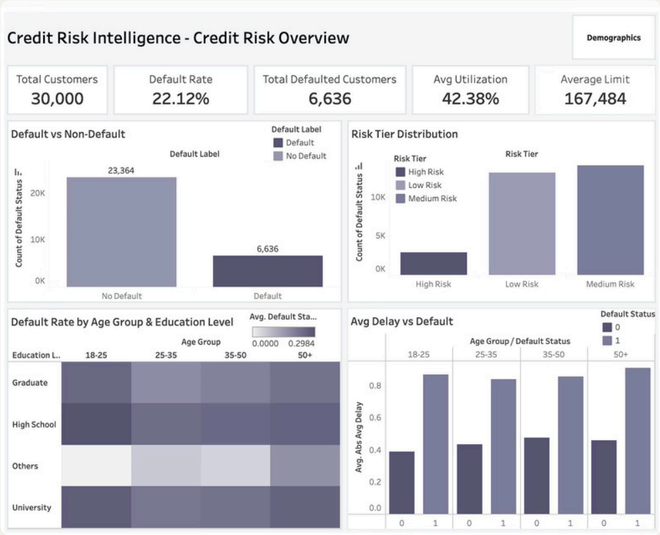
- Segmented customers intorisk tiers using behavioral features like avg_delay and utilization.
- Identified key drivers of default through root cause analysis and validated themusing hypothesis testing.

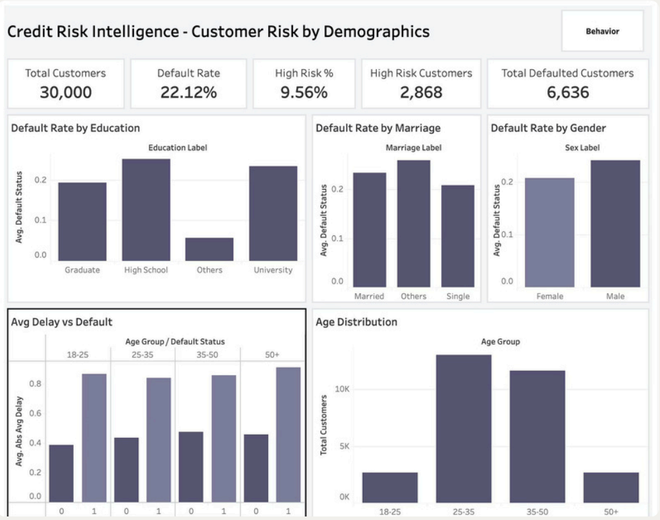
Python Libraries & Methods

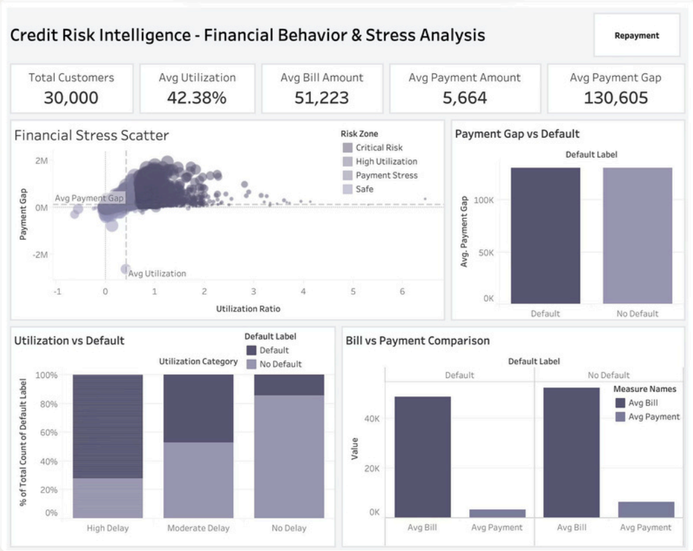
- Usedpandas, numpy, seabornfor data processing and visualization.
- Applied scipy, sklearn, statsmodels for testing, segmentation, and modeling.

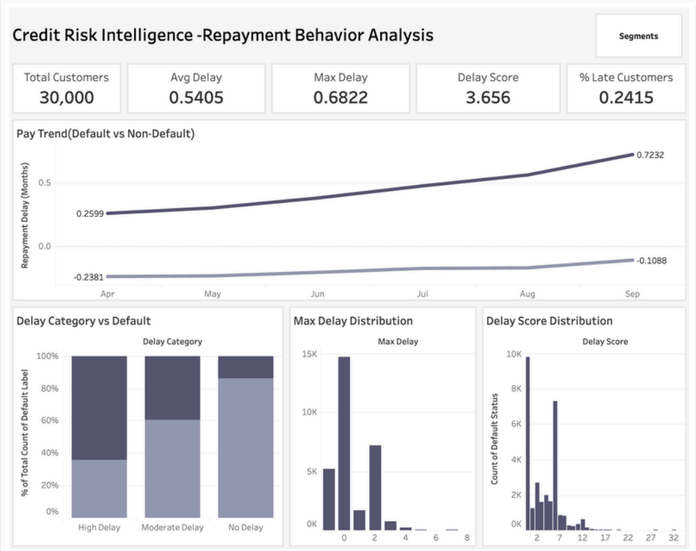
Tableau Credit Risk Intelligence Dashboard











Recommendations

Someclear, specific,and actionablebusinessrecommendations. Each must link directly to an insight. Format: Insight→Recommendation → Expected Impact.

Recommendation 1

High avg_delay strongly predicts default → **Implement early warning system for customers with delay >1 months** → Reduces defaults by identifying high-risk customers 2–3 months in advance

Recommendation 2

Low credit limit customers default more due to financial stress → **Provide controlled credit limit increases for reliable medium-risk customers** → Improves repayment capacity and reduces stress-driven defaults

Recommendation 3

High utilization signals financial distress → **Monitor and flag customers with high utilization (>80%) for proactive outreach** → Prevents escalation into default and improves portfolio stability

Impact & Value

Quantified Estimates

- ~10–15% reduction in default rate → ~$1M+ annual revenue retained (based on lowering losses from high-risk customers).
- Early intervention reduces late-stage collections → ~20– 25% cost savings in recovery operations.
- Improved risk segmentation → ~15% increase in efficient credit allocation and lower bad debt exposure

Cost of In action

Without these interventions, high-risk customers remain unidentified, leading to increasing defaults, higher recovery costs, and significant revenue loss over time.

Impact Metric 1

~10–15% reduction in defaults → Estimated $1M+ revenue retained annually (based on reducing default losses across the portfolio)

Impact Metric 2

Early risk detection & monitoring → Estimated 20–25% reduction in collection costs annually (fewer late-stage recoveries needed)

Limitations & Next Steps

Limitations

- Dataset is limited to a short time period (6 months), which may not capture long-term behavior.
- Lack of external factors (income, employment, macroeconomic data) may limit accuracy.
- Assumes past behavior fully represents future risk, which may not always hold true.

Additional Data & Anal ysis

- Includeincome, jobstatus, and credit history for better risk prediction.
- Add time-series analysis to track behavior changes over longer per iods.
- Use advanced models (e.g., Random Forest, XGBoost) for improved accuracy.

Production Roadmap

- Build a **real-time default prediction model** integrated with banking systems.
- Develop a **live risk monitoring dashboard** for continuous tracking.
- Implement an **automated alert system** for high-risk customers.